uC

U2
nice_nano_clean

GND 26	GND	BAT	25
P0_6 1	TX0/P0.06	BAT	24
P0_8 2	RX1/P0.08	GND	23 GND
GND 3	GND	RST	22 MESH_RST
GND 4	GND	VCC	21 3V3
5	P0.17	AIN7/P0.31	20
6	P0.20	AIN5/P0.29	19 DET_2V5
LED_ENABLE 7	P0.22	AIN0/P0.02	18 MISO
LED_DRIVE 8	P0.24	P1.15	17 MOSI
BUTTON_A 9	P1.00	P1.13	16
SCL 10	P0.11	P1.11	15 SCK
SDA 11	P1.04	NFC2/P0.10	14
P1_06 12	P1.06	NFC1/P0.09	13 ADS_ALERT

3V3 — R14 R — R17 R — RESET1 SW_Push — MESH_RST — GND

3V3 — R19 R — R20 R — B1 SW_Push — BUTTON_A — GND

3V3 — R21 R — R22 R — B2 SW_Push — P0_6 — GND

3V3 — R23 R — R24 R — B3 SW_Push — P0_8 — GND

DETECTOR (JST PH 6)

J1 — DETECTOR

5V	1
DET_2V5	2
DET_LIGHT_TIGHT	3
DET_RECT	4
DET_DETECT	5
	6

EMITTER (JST PH 6)

J2 — EMITTER

EMIT_LED_CURRENT	1
LED_ENABLE	2
EMIT_LED_DRIVE	3
V_LED	4
5V	5
	6

J10 Micro_SD_Card — microSD

	1	DAT2
P1_06	2	DAT3/CD
MOSI	3	CMD
3V3	4	VDD
SCK	5	CLK
GND	6	VSS
MISO	7	DAT0
	8	DAT1
SHIELD	9	GND

ST7789 TFT 1.54 IN

U5

Backlight	1	BL
P0_8	2	CS
P0_6	3	DC
MESH_RST	4	RST
MOSI	5	MOSI
SCK	6	SCK
3V3	7	VCC
GND	8	GND

Backlight on/off

3V3 — 1 — Backlight — 2 — 3 — TFT_BL1 Conn_01x03_Socket

R9 R — R10 R — P1_06 — 3V3

H2 MountingHole

H3 MountingHole

H1 MountingHole

H4 MountingHole

R3 0 — LED_DRIVE — EMIT_LED_DRIVE

100k R4 — GND

C6 1n — GND

R3/C6: fit R3 = 10k and C6 to turn PWM on LED_DRIVE into a DC setpoint.

TEST POINTS

GND	TP1 Keystone5011	DET_LIGHT_TIGHT	TP7 Keystone5014
3V3	TP2 Keystone5010	DET_2V5	TP8 Keystone5014
5V	TP3 Keystone5010	EMIT_LED_CURRENT	TP9 Keystone5014
V_LED	TP4 Keystone5010	EMIT_LED_DRIVE	TP10 Keystone5014
DET_DETECT	TP5 Keystone5014	LED_ENABLE	TP11 Keystone5014
DET_RECT	TP6 Keystone5014		

PWR_FLAG — 3V3

PWR_FLAG — GND

ADC: ADS1115, address 0x48 (ADDR to GND)

DET_DETECT — R5 4.7k — C7 100n — GND

DET_RECT — R6 4.7k — C8 100n — GND

DET_LIGHT_TIGHT — R7 4.7k — C9 100n — GND

EMIT_LED_CURRENT — R8 4.7k — C10 100n — GND

3V3 — C11 100n — GND

U3 ADS1115IDGS

	8	VDD	
	2	ALERT/RDY	
AIN0	4		
AIN1	5		
AIN2	6	SCL	10
AIN3	7	SDA	9
GND	3	ADDR	1

3V3 — R11 10k — ADS_ALERT

3V3 — R2 4.7k — SCL

3V3 — R1 4.7k — SDA

i2c pullup

UV-Control v0.4: restarted from Rook v0.4 (Pro Micro, display, buttons, microSD), plus the ADS1115,
the detector/emitter JST connectors and test points. Earlier draft: hardware/uv-control/reference/v0.4-draft/.

5V and V_LED (the UV LED supply) come from the new boost, still to be added.

Spare Pro Micro pins: P0.17, P0.20, P0.10, P1.13, P0.31 (AIN7, e.g. to measure V_LED).

LED_ENABLE: on the emitter, HIGH turns the LED off (it grounds the current setpoint).
LOW or floating lets LED_DRIVE set the current (3.3 V -> ~26 mA).

J1/J2 are both 6-pin JST PH: use pin-1-to-pin-1 cables and never swap them
(the emitter port puts V_LED and 5V on detector outputs).

R5-R8 + C7-C10: ADC input protection and low-pass (4.7k/100n, about 340 Hz).

R1/R2: I2C pull-ups. R11: ALERT/RDY is open-drain. C11 at U3 VDD.

License: CERN-OHL-S-2.0
https://github.com/edgecollective/uv254

Edge Collective

Sheet: /
File: uv-control.kicad_sch

Title: UV-Control

Size: A3	Date: 2026-11-13	**Rev: v0.4**
KiCad E.D.A. 10.0.8		Id: 1/1